Roll No.: .....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech

Name of the Course: Programming in Java

Time: 3 Hours

Semester: IV

Course Code: TCS-408

Maximum Marks: 100

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Describe the architectural relationship between the JDK, JRE, and the JVM along with its key components. Explain in detail how Java achieves platform independence and describe the role of the Java Virtual Machine (JVM) and bytecode in this process.	CO1

(b)	Define a class Complex to represent an object for a complex number like Z = X + iY with the following methods:
	a) Complex add (Complex z1, Complex z2) //To add two complex numbers
	b) Complex sub (Complex z1, Complex z2) //To subtract two complex numbers
	c) Complex mul (Complex z1, Complex z2) //To multiply two complex numbers
	d) float magnitude (Complex z)//To find the modulus
	e) Complex conjugate (Complex z)//To find the complex conjugate
	
	Write a java program to demonstrate the functionality of Complex class by performing various operations on user-defined complex numbers and displaying the results in a clear and readable format.
		CO1

(c)	What is a String literal in Java? How are String literals handled in the Java Virtual Machine (JVM), specifically in the String pool? Explain the purpose and behaviour of the following String methods with a programming example. a) compareTo(String anotherString). b) equals(Object anObject) vs. equalsIgnoreCase(String anotherString) c) replace() method	CO1
Q2		
(a)	i. Explain the role of the final keyword in the context of inheritance. How does using final with a class, method, or variable affect inheritance? ii. Discuss key differences between abstract class and interface in Java.	CO2
(b)	You are developing a notification system. You have interfaces EmailNotifier and SMSNotifier, each with a default method sendNotification(String message). You need to create a class CombinedNotifier that implements both interfaces. Describe the conflict that arises when calling sendNotification(String message) on a	CO2

	CombinedNotifier object. Implement a solution that allows the CombinedNotifier to send notifications via both email and SMS and explain how you handled the ambiguity.	
(c)	Write a Java program that models a circle and allows its position to be shifted on a 2D plane. Implement this using inheritance, where a base class Shape defines the fundamental properties of any geometric shape (like position), and a derived class Circle extends Shape to include circle-specific attributes (like radius). The Circle class should have a method to shift its position by a given displacement, utilizing the super keyword to access and modify the position properties inherited from the Shape class.	CO2
Q3		
(a)	What is the exception hierarchy in Java? Explain the difference between a checked exception and an unchecked exception in Java. Provide an example of each.	CO3
(b)	Create a Java program that reads a file containing a list of names, one name per line. Sort the names alphabetically and write the sorted list to a new file named sorted_names.txt. Handle potential FileNotFoundException and IOException.	CO3
(c)	What are the different states in the lifecycle of a Java thread? Create a Java program where multiple threads try to increment a shared counter variable. Implement proper synchronization to ensure that the final value of the counter is correct, even with multiple threads accessing it concurrently.	CO3
Q4		
(a)	Explain the need for Generic classes and Generic methods in Java. Write a generic method swap<T>(T[] arr, int i, int j) that swaps the elements of an array of different types (e.g. Integer, String) at the given indices in the array.	CO5
(b)	WAP in Java to manage a simple list of student names using an ArrayList. Implement the following functionalities: 1. Create (Add): Add a new student name to the list. 2. Read (Retrieve): Retrieve and print all student names in the list. Retrieve and print the student's name at a specific index. Handle the case where the index is out of bounds. 3. Update (Modify): Update the student's name at a specific index with a new name. Handle the case where the index is out of bounds. 4. Delete (Remove): Remove a student name from the list at a specific index. Handle the case where the index is out of bounds.	CO5
(c)	Create a Swing application with a JTextArea and a JButton labeled "Count Words". When the button is clicked, count the number of words in the JTextArea and display the count in another JLabel. (Assume words are separated by spaces).	CO4
Q5		
(a)	Develop a Swing application with a JTextField. Implement a KeyListener such that every character typed in the text field is immediately converted to uppercase and displayed in the same text field.	CO4
(b)	Design a Swing application with a JPanel. Implement a MouseListener such that when the user clicks anywhere on the panel, the coordinates of the click (x, y) are displayed in a JLabel below the panel.	CO6
(c)	Write a Java code snippet that executes a simple SELECT query (e.g., SELECT * FROM employees) on a database using a Statement object. Iterate through the ResultSet and print the values of a specific column (e.g., emp_code, emp_name, emp_salary). Remember to close the Statement and ResultSet in a finally block.	CO6